Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

[10pt, a4paper]article

[review]daal-2024

Title of the DAAL 2024 Paper (Title in 14-point Bold)

Author1, Author2, Author3

Affiliation1, Affiliation2, Affiliation3
Address1, Address2, Address3
author1@xxx.yy, author2@zzz.edu, author3@hhh.com
{author1, author5, author9}@abc.org

Ringkasan

Each paper must include an abstract of 150 to 200 words in 9 pt with interlinear spacing of 10 pt. The heading Abstract should be centered, 10 pt bold.

keyword1, keyword2, keyword3

1 Full Paper Submission

DAAL 2024 has both long and short papers featuring substantial, original, and unpublished research in all aspects of data-driven approaches to historically attested languages. This includes, but is not limited to language resources, data quality and digitisation issues.

Submissions may be of two types:

- Regular long papers - up to eight (8) pages maximum,* presenting substantial, original, completed, and unpublished work.
- Short papers - up to four (4) pages,[1] describing a small, focused contribution, negative results, system demonstrations, etc.

Upon acceptance, final versions of long papers will be given one additional page – up to nine (9) pages of content plus unlimited pages for acknowledgements and references – so that reviewers' comments can be considered. Final versions of short papers may have up to five (5) pages, plus unlimited pages for acknowledgements and references. All figures and tables that are part of the main text must fit within these page limits for long and short papers.

Papers must be of original, previously-unpublished work. Papers must be **anonymized to support double-blind reviewing**. Submissions, thus, must not include authors' names and affiliations. The submissions should also avoid links to non-anonymized repositories: the code should be either submitted as supplementary material in the final version of the paper or as a link to an anonymized repository (e.g., Anonymous GitHub or Anonym Share). Papers that do not conform to these requirements will be rejected without review.

Please use the `[review]` setting for submissions:

```
\usepackage[review]{daal-2024}
```

This hides the authors and adds page numbers.

2 How to Produce the `.pdf`

In order to generate a PDF file out of the LaTeX file herein, when citing language resources, the following steps need to be performed:

1. `xelatex your_paper_name.tex`

[1] Excluding any number of additional pages for references, ethical consideration, conflict-of-interest, as well as data and code availability statements.

2. `bibtex your_paper_name.aux`
3. `bibtex languageresource.aux` *NEW*
4. `xelatex your_paper_name.tex`
5. `xelatex your_paper_name.tex`

From 2024 we are using a sans-serif font TeX Gyre Heros, so you must install it if you do not have it. For monospaced font, we use TeX Gyre Cursor.[2] To compile, you can use either `pdfLaTeX`, `XeLaTeX` or `LuaLaTeX`.

3 Final Paper

Each final paper should be submitted to colin.swaelens@ugent.be. The fully justified text should be formatted according to DAAL-2024 style as indicated for the Full Paper submission.

Papers must be up to 5 pages for short papers and 9 pages for long papers, including figures (plus more pages if needed for references, ethical consideration, conflict-of-interest, as well as data and code availability statements). Length does not depend on the presentation mode (oral or poster).

- The paper is in A4-size format, that is 21 x 29.7 cm.
- The text height is 24.7 cm and the text width 16.0 cm in two columns separated by a 0.6 cm space.
- The font for the entire paper should be Tex Gyre Heros, a sans-serif font based Helvetica, with TeX Gyre Cursor for the monospaced font.
- The main text should be 10 pt with an interlinear spacing of 11 pt.
- The use of `daal-2024.sty` will ensure the good formatting.

3.1 General Instructions for the Final Paper

The unprotected PDF files will appear in the online proceedings directly as received. **Do not print the page number**.

4 Page Numbering

Please do not include page numbers in your Paper. The definitive page numbering of papers published in the proceedings will be decided by the Editorial Committee.

5 Headings / Level 1 Headings

Level 1 Headings should be capitalised in the same way as the main title, and centered within the column. The font used is 12 pt bold. There should also be a space of 12 pt between the title and the preceding section and 3 pt between the title and the following text.

[2]They are available from CTAN in the tex-gyre-heros and tex-gyre-cursor packages or in TeXLive as `tex-gyre` and `tex-cursor`.

Output	natbib command	Old command
Eco:1990	`\citep`	`\cite`
Eco:1990	`\citealp`	no equivalent
Eco:1990	`\citet`	`\newcite`
Eco:1990	`\citeyearpar`	`\shortcite`

Tabel 1: Citation commands supported by the style file. The style is based on the natbib package and supports all natbib citation commands. It also supports commands defined in previous style files for compatibility.

5.1 Level 2 Headings

The format of Level 2 Headings is the same as for Level 1 Headings, with the font 11 pt, and the heading is justified to the left of the column. There should also be a space of 6 pt between the title and the preceding section and 3 pt between the title and the following text.

5.1.1 Level 3 Headings

The format of Level 3 Headings is the same as Level 2, except that the font is 10 pt, and there should be no space left between the heading and the text as in 5.1.1. There should also be a space of 6 pt between the title and the preceding section and 3 pt between the title and the following text.

6 Citing References in the Text

6.1 Bibliographical References

Table 1 shows the syntax supported by the style files. We encourage you to use the natbib styles. You can use the command `\citet` (cite in text) to get "author (year)" citations, like this citation to a paper by CastorPollux-92. You can use the command `\citep` (cite in parentheses) to get "(author, year)" citations CastorPollux-92. You can use the command `\citealp` (alternative cite without parentheses) to get "author, year" citations, which is useful for using citations within parentheses (e.g. CastorPollux-92).

When several authors are cited, those references should be separated with a semicolon: [**?**, **?**]. When the reference has more than three authors, only cite the name of the first author followed by "et. al.", e.g. Superman-Batman-Catwoman-Spiderman-00.

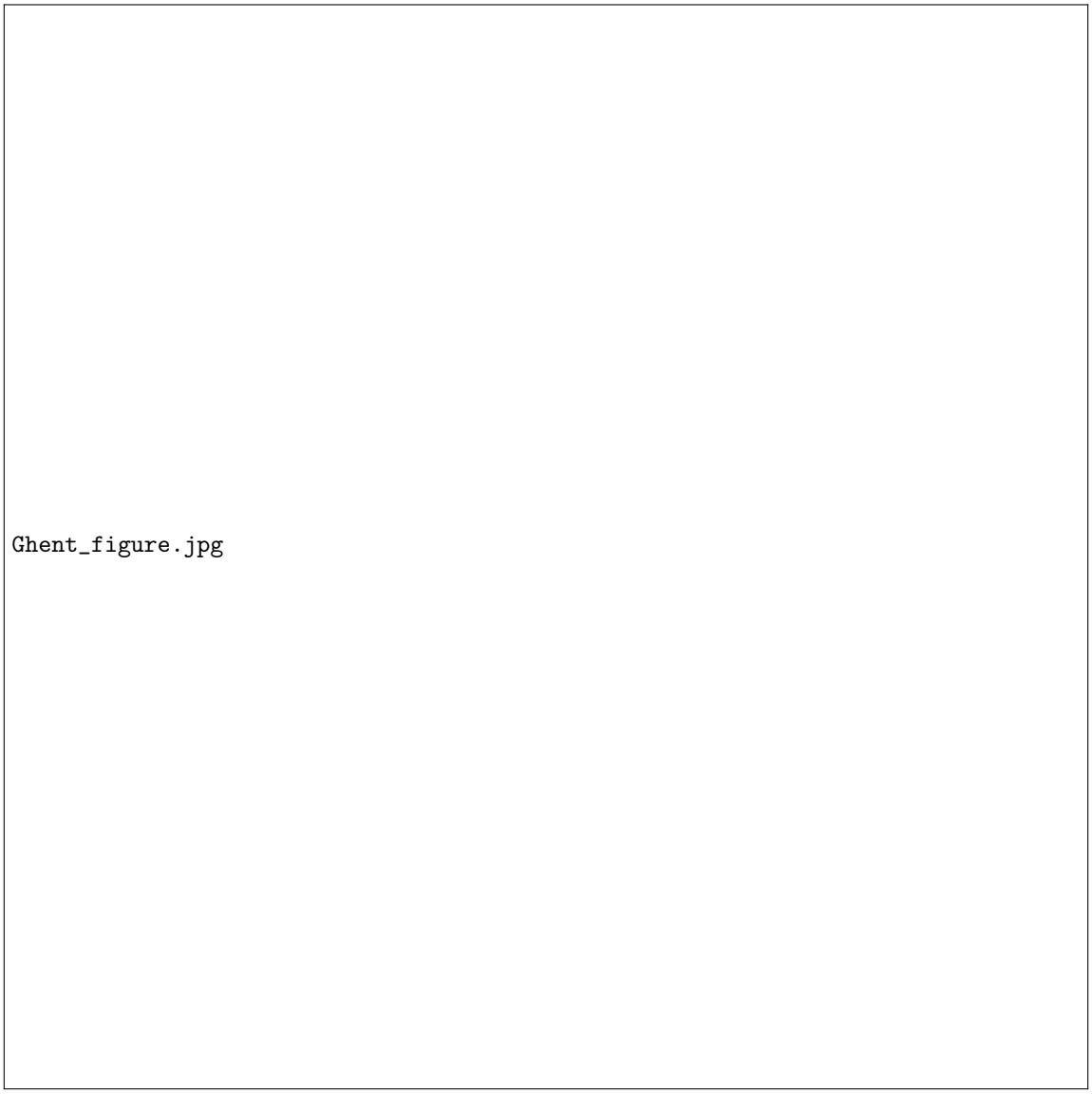
7 Figures & Tables

7.1 Figures

All figures should be centred and clearly distinguishable. They should never be drawn by hand, and the lines must be very dark in order to ensure a high-quality printed version. Figures should be numbered in the text, and have a caption in 10 pt underneath. A space must be left between each figure and its respective caption.

Example of a figure:

Figure and caption should always appear together on the same page. Large figures can be centered, using a full page.



Gambar 1: The caption of the figure.

7.2 Tables

The instructions for tables are the same as for figures.

l|X|
Level|Tools
Morphology | Pitrat Analyser
Syntax | LFG Analyser (C-Structure)
Semantics | LFG F-Structures + Sowa's Conceptual Graphs

Tabel 2: The caption of the table

8 Footnotes

Footnotes are indicated within the text by a number in superscript[3].

9 Copyrights

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10 Conclusion

Your submission of a finalised contribution for inclusion in the DAAL Proceedings automatically assigns the above copyright to the organisation.

11 Acknowledgements

Place all (including those concerning research grants and funding) in a separate section at the end of the paper.

12 Optional Supplementary Materials

Appendices or supplementary material (software and data) will be allowed ONLY in the final, camera-ready version, but not during submission, as papers should be reviewed without the need to refer to any supplementary materials.

Each **camera ready** submission can be accompanied by an appendix usually being included in a main PDF paper file, one `.tgz` or `.zip` archive containing software, and one `.tgz` or `.zip` archive containing data.

We encourage the submission of these supplementary materials to improve the reproducibility of results and to enable authors to provide additional information that does not fit in the paper. For example, preprocessing decisions, model parameters, feature templates, lengthy proofs or derivations, pseudocode, sample system inputs/outputs, and other details necessary for the exact replication of the work described in the paper can be put into the appendix. However, the paper submissions must remain fully self-contained, as these supplementary materials

[3]Footnotes should be in 9 pt, and appear at the bottom of the same page as their corresponding number. Footnotes should also be separated from the rest of the text by a 5 cm long horizontal line.

are optional, and reviewers are not even asked to review or download them. If the pseudo-code or derivations, or model specifications are an essential part of the contribution, or if they are important for the reviewers to assess the technical correctness of the work, they should be a part of the main paper and not appear in the appendix.

12.1 Appendices

Appendices are material that can be read and include lemmas, formulas, proofs, and tables that are not critical to the reading and understanding of the paper. It is highly recommended that the appendices should come after the references; the main text and appendices should be contained in a 'single' manuscript file, without being separately maintained. Letter them in sequence and provide an informative title: *Appendix A. Title of Appendix*

12.2 Extra space for ethical considerations and limitations

Please note that extra space is allowed after the 8th page (4th page for short papers) for an ethics/broader impact statement and a discussion of limitations. At submission time, if you need extra space for these sections, it should be placed after the conclusion so that it is possible to rapidly check that the rest of the paper still fits in 8 pages (4 pages for short papers). Ethical considerations sections, limitations, acknowledgements, and references do not count against these limits. For camera-ready versions, nine pages of content will be allowed for long (5 for short) papers.

13 Providing References

13.1 Bibliographical References

Bibliographical references should be listed in alphabetical order at the end of the paper. The title of the section, "Bibliographical References", should be a Level 1 Heading. The first line of each bibliographical reference should be justified to the left of the column, and the rest of the entry should be indented by 0.35 cm.

The examples provided in Section 14 (some of which are fictitious references) illustrate the basic format required for papers in conference proceedings, books, journal articles, PhD theses, and books chapters.

14 Bibliographical References